

Active-Learning Phage Engineering for Xanthomonas Biocontrol

iGEM Claremont 2026 —Team Onboarding

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Part 0 —Roadmap

TL;DR

- **Closed-loop active learning** for phage RBP engineering against *Xanthomonas campestris* pv. *campestris*.
- **Modules 00-07 done** (computational); Module 08 (wet-lab data) opens ~June 1.
- First Boltz-2 result on rbp_01 (712 aa) × TonB: **ipTM = 0.365, chain_A_ptm = 0.808.**
- BALD recommends the next 4 variants + 1 random control in **< 1 second.**

Targeting: Best Agriculture Project · Best Model · Best Composite Part.

Roadmap

- 1 **The science** —Xanthomonas, phage host-range, the data-scarcity problem. [WET LAB] [PI]
- 2 **Pipeline architecture** —Modules 00-08, data-contract convention. [DRY LAB] [PI]
- 3 **The ML core** —deep ensemble, BALD, code + sample outputs. [DRY LAB] [PI]
- 4 **Current Boltz-2 result** —what the 0.365 ipTM means. all
- 5 **The 48-hour cycle** —wet/dry handoffs, ELISA, knockouts. all
- 6 **Reproducing & demoing** —quick-start, tests, Laguna. [DRY LAB] [WET LAB]
- 7 **Risks, decisions, asks** [PI]
- 8 **References & appendix** all

How to read this deck

Tags on each section header tell you who needs it:

- **[WET LAB]** —biological motivation, lab protocols, handoffs you'll execute.
- **[DRY LAB]** —code, algorithms, file conventions, reproducibility.
- **[PI]** —overall framing, decisions needed, risks.

Tagless sections are for everyone.

Suggested cuts

- **30-min wet-lab onboarding:** slides 1-4, 7-12, 17-22 (modules at a glance), 37-44 (the 48-hour cycle and protocols).
- **45-min PI / advisor briefing:** slides 1-4, 5-12 (science), 13-14, 23, 28-32 (ML core), 33-36 (Boltz-2), 51-55 (risks + decisions), 56 (deliverables).
- **90-min full team deep dive:** all slides.

Demo (≈ 15 min) is independent; run `DEMO.md` after slide 48 or at the end.

Part 1 —The science · [WET LAB] [PI]

Key vocabulary —for this part

- **Phage / lytic phage** —virus that infects bacteria; lytic = kills the host after replicating (what we want for biocontrol).
- **RBP (receptor-binding protein)** —the protein on a phage' s tail that latches onto a specific bacterial surface receptor. The “lock-and-key” step.
- **Pathovar (pv.)** —a strain of a plant-pathogenic bacterium defined by which plant it infects.
- **AUC** —single score from 0.5 (random guess) to 1.0 (perfect) summarising binary-classification quality.
- **BLAST** —sequence-similarity search. Misses proteins that have diverged so much that the letters no longer match.
- **HMM (Hidden Markov Model)** —sequence-profile search that catches what BLAST misses by learning a family' s conserved pattern.
- **TonB / ExbB / ExbD1** —bacterial outer-membrane import complex; phiL7' s entry point (Hung 2003).

Xanthomonas —what & why we care

- Genus of plant-pathogenic gamma-proteobacteria; >400 host plant species (Ryan 2011, Nat Rev Microbiol).
- California-relevant pathovars:
 - **Xcc** —black rot of brassicas (cabbage, broccoli, kale).
 - *X. citri* subsp. *citri* —citrus canker.
 - *X. perforans* / *X. euvesicatoria* —bacterial spot of tomato.
- Current control: **copper bactericides**; resistance now widespread (Aiello 2019, Plant Dis).
- Field trials show phage biocontrol matches copper efficacy (Iriarte 2018; Holtappels 2022).

Phage biocontrol —the host-range problem

- Phages are **host-specific** —that' s the feature (no off-target microbes) AND the bug (each strain needs the right phage).
- Predicting which phage infects which bacterium is unreliable:
 - **PhageHostLearn** (Boeckaerts 2024, Nat Commun) reaches **AUC 0.82 within genus**.
 - **Cross-genus AUC drops to 0.67-0.70** (PAML benchmark, Mutalik 2025).
- The bottleneck: labeled (phage, host) interaction data is scarce, expensive, species-bound.

Our reference scaffold

Host	Xcc ATCC 33913	NCBI: GCF_000007145.1
Phage	phiL7	NCBI: EU717894
Source	da Silva 2002 (Nature); Lee 2009 (AEM)	

- Dry-lab uses these public references.
- **Wet lab self-isolates** from California brassicas —bypasses 4-month USDA APHIS PPQ-526 permit (per PI consultation 2026-05-07).

phiL7 receptor system —Hung 2003

Hung, C.-H. et al. (2003) BBRC 302:878-884, PMID 12646254.

- phiL7 enters Xcc through the **TonB-ExbB-ExbD1** complex (energy-coupled outer-membrane import).
- Tn5 mutagenesis + complementation: **TonB, ExbB, ExbD1 essential.**
- Δ exbD2 (strain CH620) retains full sensitivity → **ExbD2 is NOT required.**

Free negative control. The Hung 2003 audit (May 2026) corrected an earlier mis-citation that listed ExbD2 as essential. Building Δ exbD2 alongside Δ tonB / Δ exbB / Δ exbD1 validates the entire knockout system: Δ exbD2 should still allow infection.

phiL7' s RBP —Lee 2009 and our HMM rediscovery

Lee, C.N. et al. (2009) AEM 75:7828 —the phiL7 genome paper.

- Lee 2009 **explicitly searched** for an OP1 tail-fiber (ORF25) homolog and **found none** by sequence similarity:
“We were unable to identify a homolog of the OP1 tail fiber protein (ORF25)…”
- p20 (1105 aa) was suggested as host-range related; no protein was named a “tail spike” .
- **Our rbp_01 (712 aa)** comes from PhageRBPdetect' s Tail_spike_N HMM —a structural-profile method that finds proteins too diverged for BLAST.
- **Not a contradiction with Lee 2009 —a complementary discovery using a more sensitive tool.**

The data-scarcity bottleneck

- We have **one** experimentally confirmed (phage, host) interaction for our system: phiL7 × Xcc.
- The interaction matrix (Module 01) has 2,236 phage-host pairs from literature curation —none are quantitative binding affinities.
- For a 712 aa protein, the variant space is 20^{712} —astronomically larger than any wet-lab budget.
- **We need a learning strategy that gets the most out of each expensive ELISA measurement.**

Why active learning is the right framing

- **Active learning (AL):** the model picks its next training point by maximizing expected information gain (Lindley 1956; Settles 2009).
- Demonstrated 2-5 $\times$ speedup over random selection in adjacent domains:
 - **Hie 2024** (Nat Biotechnol) —human antibody affinity maturation with ESM-1b/1v + AL (~20 variants per antibody).
 - **Yang 2025 ALDE** (Nat Commun) —DNN ensemble + **Thompson sampling** (not BALD) on enzyme directed evolution, 12% $\rightarrow$ 93% yield in 2 rounds.
- **None applied to phage RBP $\times$ bacterial receptor.** That's our methodological contribution.

What this maps to in iGEM

Track	Our deliverable
Best Agriculture Project	Phage biocontrol pipeline against Xcc
Best Model	Closed-loop AL + UQ + causal validation
Best Composite Part	Registered RBP-His6 expression library

Plus: in-house Xanthomonas + phage isolates, sequenced + deposited.

Part 2 —Pipeline architecture · [DRY LAB] [PI]

Key vocabulary —for this part

- **Module** —a numbered subdirectory (`00_...` through `08_...`). Each does one job; outputs feed the next.
- **ORF (open reading frame)** —a stretch of DNA coding for a single protein (start codon → stop codon).
- **Gitignored** —listed in `.gitignore` so git won't track it. Large binaries (genomes, weights) are excluded but recorded in `MANIFEST.csv` so the set stays reproducible.
- **MANIFEST.csv** —per-folder ledger (filename · SHA-256 · size · record count) that makes gitignored outputs verifiable.
- **Conda env** —project-specific Python install with every library pinned to an exact version.
`conda activate igem2026`.
- **Notebook-first** —author exploratory code as `.ipynb`; freeze stable logic to `.py`.
- **Bilingual comments** —every code comment in this repo is written in English AND Chinese.

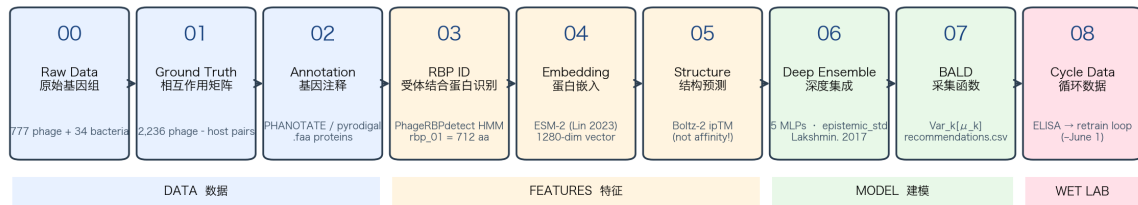
The three layers

Layer 0 —Priors	Boltz-2 ipTM, ESM-2, PLM-interact (optional)
Layer 1 —AL loop	Ensemble → BALD → ELISA → retrain
Layer 2 —Causality	Δ_{tonB} / Δ_{exbB} / Δ_{exbD1} knockouts

Binding (Layer 1) is necessary but not sufficient for infection (Farquharson 2021).
 Layer 2 decouples receptor-specific binding from defense-system contributions.

Pipeline at a glance —Modules 00 → 08

iGEM Claremont 2026 — Active-Learning Pipeline / 主动学习流程



Color coding: data (blue) · features (yellow) · ML (green) · wet lab (pink).

The data-contract convention

Module N — data contract / 模块内数据流约定



outputs/ → Module (N+1) inputs/ (READ-ONLY POINTER, no copy)

■ inputs/ = pointers to upstream outputs (read-only).

■ processes/ = the **only** place code lives.

■ outputs/ = canonical artifacts + MANIFEST.csv (large files gitignored)

Notebook-first workflow + bilingual comments (CLAUDE.md)

- New code is authored as **Jupyter notebooks first** (<NN>_<short_name>.ipynb).
- Once a notebook (a) runs end-to-end and (b) matches a verification cell → **freeze to .py**.
- **Bilingual comments** mandatory —English / 中文 on the same line for short comments.
- Module 07 is the exception —production orchestration, written as .py from day 1.
- nbstripout enforced repo-wide so notebook outputs never pollute git diffs.

Module 00 —Raw Data

- **777 phage + 34 bacteria** complete genomes from NCBI.
- Genome binaries **gitignored** (~630 MB); re-fetchable via:
 - `python 00_raw_data/processes/fetch_phages.py`
 - `python 00_raw_data/processes/fetch_bacteria.py`
- **MANIFEST.csv** carries SHA-256 + size per genome for reproducibility.
- Canonical reference pair: **EU717894.1.fna** (phiL7) + **GCF_000007145.1.fna** (Xcc).

Module 01 —Ground Truth interaction matrix

01_data_ground_truth/outputs/interaction_matrix.csv —first 4 rows:

phage_acc	host_acc	label	source	notes
EU717894.1	GCF_000007145.1	1	literature_curated	Hung 2003 (receptor); Lee 2009 (genome)
NC_054459.1	NZ_CP150073	1	literature_curated	X. oryzae pv. oryzae
ON758385.1	—	1	literature_curated	Xanthomonas sp.
ON711490.1	—	1	literature_curated	Xcc XC114

- 2,236 pairs (315 positive + 1,921 negative + 1 ground truth).
- 22/22 tests pass.

Module 02 —Annotation

- **Phage:** PHANOTATE (McNair 2019) —dynamic-programming over overlapping ORFs.
- **Bacteria:** pyrodigal binding of Prodigal (Hyatt 2010).
- **Never swap them.** Prodigal assumes non-overlapping ORFs → loses ~10-15 % of phage genes.
- Optional second pass: pharokka (PHROG / VFDB / CARD) for functional categories.

Genome	Tool	ORFs found
phiL7 (EU717894.1)	PHANOTATE	80
Xcc (GCF_000007145.1)	pyrodigal	4,344

Module 03 —RBP identification (PhageRBPdetect, Boeckaerts 2022)

03_rbp_identification/outputs/EU717894.1_rbp_candidates.csv —top 3 of 80:

orf_id	length_aa	hmm_score	hmm_match	combined_score	rank
EU717894.1_orf_00001	712	342.0	unknown_C54	1.000	1
EU717894.1_orf_00021	918	235.1	unknown_C112	1.000	2
EU717894.1_orf_00003	224	56.7	unknown_C294	1.000	3

- **rbp_01** (= orf_00001, 712 aa, HMM score 342) —primary Cycle 0 target.
- Hits the Tail_spike_N HMM —exactly what Lee 2009 couldn't find via BLAST.
- 25+ tests pass (2 expected fails until you run hmmpress locally).

Module 04 —Protein embedding (ESM-2, Lin 2023)

- ESM-2 = masked-language-model trained on ~65 M protein sequences (UniRef50/90).
- Mean-pooling over residues → sequence-level vector.

Variant	Dim	Where it runs
esm2_t6_8M_UR50D	320	Local CPU (proof of concept; current outputs)
esm2_t33_650M_UR50D	1280	Laguna A100 / L40S (production target)
esm2_t36_3B_UR50D	2560	Laguna only (final benchmark)

Optional layer: PLM-interact (Liu 2025) —ESM-2 fine-tuned on human PPIs. AUPR +16-28 % when transferred to mouse / fly / worm / yeast / E. coli PPI. **Untested on phage-bacteria** —our project may be the first.

Module 05 —Structure prediction (Boltz-2 + AF3)

- **Boltz-2** (Passaro 2025) —predicts complex 3D structure + ipTM interface confidence.
- **AF3** (Abramson 2024) —higher-quality static structures; trimer support.

Critical caveat —Boltz-2 affinity head is small-molecule only. It was trained on PubChem / ChEMBL / BindingDB. For **protein - protein pairs** (e.g., RBP × TonB) the affinity head outputs **NaN**. Use **ipTM** as a structural confidence proxy, NOT a quantitative binding affinity.

Part 3 —The ML core · [DRY LAB] [PI]

Key vocabulary —neural-net basics

- **Neural network** —a chain of mathematical functions arranged in “layers” that turns one number-vector into another. Trained by adjusting the numbers inside each layer to minimise a loss on training examples.
- **MLP (multilayer perceptron)** —the simplest architecture: fully-connected layers separated by a nonlinearity. What our Module 06 uses.
- **Layer / hidden layer** —one matrix multiplication + offset. “Hidden” = intermediate representation, not the final prediction. Our MLP has 3 hidden layers of widths 256 / 256 / 128.
- **ReLU** —keeps positive numbers unchanged, replaces negatives with zero. Without nonlinearity, stacked layers collapse to one linear function.
- **Dropout** —randomly turn off connections each training step. Forces the network to spread information rather than memorise.
- **Gaussian** —bell curve; described by mean (centre) and σ (spread). Our model outputs (μ, σ) per input.
- **NLL loss** —penalises low predicted probability on the truth. Wrong-but-uncertain hurts less than wrong-and-confident.

Key vocabulary —uncertainty + active learning

- **Deep ensemble** —train the same architecture multiple times with different random initialisations. Members agree where data is informative; disagree where it isn't.
- **Epistemic vs aleatoric** —epistemic = model ignorance, shrinks with more data. Aleatoric = irreducible measurement noise (ELISA pipetting variance). BALD targets epistemic only.
- **Calibration / ECE** —does a “90 % credible interval” actually contain the truth 90 % of the time? Expected Calibration Error measures the gap.
- **Mutual information** — “how much does knowing X tell me about Y ?” Zero if they're independent.
- **BALD** —picks the experiment whose result gives the most mutual information about the model's parameters.
- **Acquisition function** —any rule that ranks unmeasured candidates and recommends which to test next. BALD is one; Thompson sampling (ALDE) is another.

Module 06 —Deep ensemble for predictive uncertainty

- **Lakshminarayanan 2017** (NeurIPS) —5 MLPs, independently trained, predictive mean + uncertainty.
- Each member predicts a Gaussian (μ_k, σ_k^2) via Gaussian NLL loss.

Why deep ensembles? - Better calibration than MC Dropout (Ovadia 2019). - Scales to ESM-2' s 1280-D inputs (Gaussian Processes don' t).

Greenman 2025 audit: ” There is no single best UQ method” across protein-engineering benchmarks. Ensembles tend to be most accurate but worst-calibrated. We pick ensembles because (a) scalability, (b) ALDE' s precedent, (c) ECE can be patched with temperature scaling.

ensemble.py —per-member prediction (snippet)

```
def forward(self, x: torch.Tensor) -> Tuple[torch.Tensor, torch.Tensor]:
    """
    Returns (mean, sigma) — both shape (batch,).
    返回 (均值, 标准差), 各形状为 (batch,)。
    """
    h = self.net(x)
    mean = self.head_mean(h).squeeze(-1)
    # Clamp log_sigma to [-7, 7] for numerical stability
    # 将 log_sigma 截断到 [-7, 7] 防止数值爆炸
    log_sigma = self.head_log_sigma(h).squeeze(-1).clamp(-7.0, 7.0)
    sigma = torch.exp(log_sigma)
    return mean, sigma
```

Source: 06_uncertainty_model/processes/ensemble.py:57-68.

Epistemic vs aleatoric —the key decomposition

For a Gaussian ensemble of K members, total predictive variance decomposes as:

$$\sigma_{\text{total}}^2(x) = \underbrace{\text{Var}_k[\mu_k(x)]}_{\text{epistemic (reducible)}} + \underbrace{\mathbb{E}_k[\sigma_k^2(x)]}_{\text{aleatoric (noise)}}$$

- **Epistemic** = what the model doesn't know → shrinks with more data → **BALD's target**.
- **Aleatoric** = measurement / experimental noise → fundamentally irreducible.

`predictions.csv` exports both as `std` (total) and `epistemic_std` (BALD input).

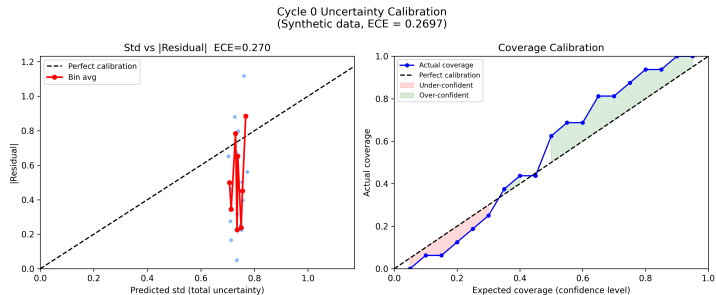
Module 06 output —Cycle 0 (synthetic)

`06_uncertainty_model/outputs/cycle_0/predictions.csv` —head:

rbp_id	receptor_id	predicted_score	std	epistemic_std
rbp_00	rec_00	4.872	0.736	0.125
rbp_00	rec_01	4.959	0.755	0.114
rbp_01	rec_02	5.128	0.726	0.190
rbp_01	rec_03	5.094	0.715	0.049
rbp_07	rec_02	5.177	0.721	0.218

- 80 (RBP × receptor) pairs scored.
- `epistemic_std` ranges 0.04-0.22 across the pool.
- `model_version` = `aa99d51_cycle_0` (git SHA + cycle).

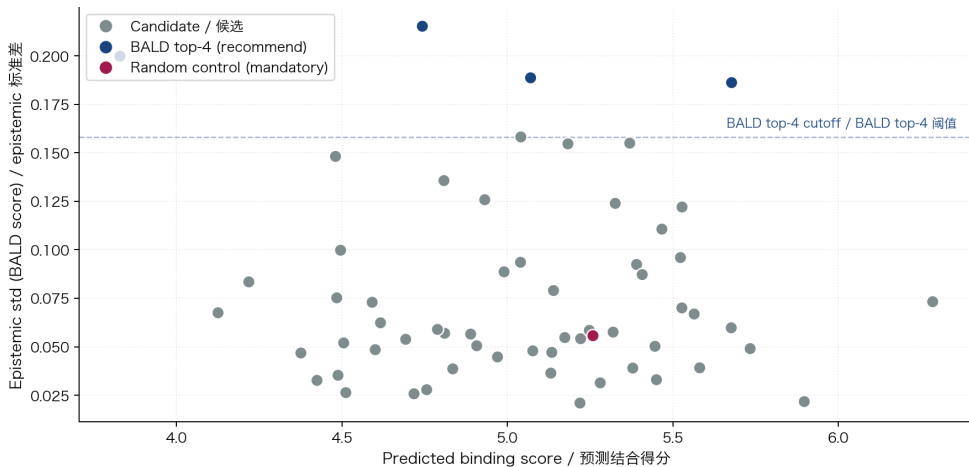
Calibration plot —Cycle 0 (synthetic)



Reliability diagram: predicted credible-interval coverage vs observed. On synthetic data, calibration is near-diagonal —a meaningful version will be plotted after Cycle 0 ELISA arrives.

Module 07 —BALD intuition

BALD selection intuition — pick where the model disagrees most
BALD 选取直观：选模型成员分歧最大的点



BALD math (regression extension)

Original (Houlsby 2011, GPC classification):

$$\text{BALD}(x) = I(y; \theta \mid x, D) = H[y \mid x, D] - \mathbb{E}_{\theta \sim p(\theta \mid D)}[H[y \mid x, \theta]]$$

For a Gaussian deep ensemble + regression target:

$$\text{BALD}(x) \approx \text{Var}_k[\mu_k(x)] \implies \text{score} = \text{epistemic_std}$$

Extension note. Houlsby 2011 originally applied BALD to a Gaussian Process Classifier. The information-theoretic objective extends naturally to deep ensemble regression, but this is our extension—not a direct citation. Mention this when presenting to academic audiences.

bald.py —bald_score() + select_batch()

```
def bald_score(epistemic_std: np.ndarray) -> np.ndarray:
    """BALD score = epistemic_std (rank-equivalent to epistemic variance)."""
    if np.any(epistemic_std < 0):
        raise ValueError("epistemic_std must be non-negative; got negatives.")
    return epistemic_std.copy()

# Inside select_batch(): rank by BALD score (descending), then take top-K
ranked = pool.sort_values(bald_col, ascending=False).reset_index(drop=True)
ranked["bald_rank"] = ranked.index + 1
bald_picks = ranked.iloc[:n_bald].copy()
bald_picks["selection_reason"] = [f"bald_top_{i+1}" for i in range(len(bald_picks))]

# Random control: sample from remaining (NOT in BALD top-K)
remaining = ranked[~ranked.index.isin(set(bald_picks.index))]
random_picks = ranked.loc[_random.Random(seed).sample(list(remaining.index), k=n_random)]
random_picks["selection_reason"] = "random_control"
```

Source: 07_acquisition_function/processes/bald.py:38-135.

Module 07 output —Cycle 1 recommendations

07_acquisition_function/outputs/cycle_1/recommendations.csv:

rbp_id	receptor_id	predicted_score	epistemic_std	bald_rank	selection_reason
rbp_07	rec_02	5.177	0.218	1	bald_top_1
rbp_03	rec_01	4.810	0.197	2	bald_top_2
rbp_05	rec_02	4.906	0.196	3	bald_top_3
rbp_01	rec_02	5.128	0.190	4	bald_top_4
rbp_03	rec_03	5.128	0.127	19	random_control

- 4 BALD picks + 1 random control (Hie 2024 control-arm pattern).
- These are **synthetic placeholders**. Real Cycle 1 = rbp_01 variants × TonB after Cycle 0 ELISA.

ALDE caveat —Yang 2025 $\neq$ BALD validation

- **Yang 2025 ALDE** (Nat Commun) is the closest published work —active learning + DNN ensemble + protein engineering.
- But: ALDE uses **Thompson sampling** as the acquisition function and **one-hot encoding** as features. Not BALD, not ESM-2.
- So ALDE validates: “AL + UQ + DNN ensemble works in protein engineering” .
- BALD specifically still needs its own citation chain (Houlsby 2011 + our extension).

Part 4 —The current Boltz-2 result

Key vocabulary —for this part

- **Structure prediction** —given an amino-acid sequence (or two), predict the protein's 3D shape —where each atom sits. Replaces months of crystallography with a “good-enough” starting point.
- **PDB file** —the standard plain-text format for atomic 3D structures (one line per atom). Open with PyMOL (free) or ChimeraX (free) to visualise.
- **ipTM** —interface predicted TM-score (0-1). Confidence about how two chains dock. ≥ 0.6 trustworthy; 0.4-0.6 ambiguous; < 0.4 the model is essentially guessing.
- **chain pTM** —TM-score for a single chain alone. Tells you the monomer is well-predicted, independent of docking. Our `chain_A_ptm = 0.808` says `rbp_01` monomer is reliable.
- **PAE (predicted aligned error)** —2D matrix. Entry $[i, j]$ = expected $\boxtimes$ error in residue j 's position if I align my prediction at residue i . Low off-diagonal = confident interface.
- **pLDDT** —per-residue confidence in the local backbone geometry. High = the model is sure about that residue's local position.

What we ran (job 59986)

Chain A	rbp_01 (712 aa tail spike candidate, EU717894.1)
Chain B	TonB (604 aa, GCF_000007145.1)
GPU	NVIDIA L40S on CARC Laguna
Runtime	~3 minutes

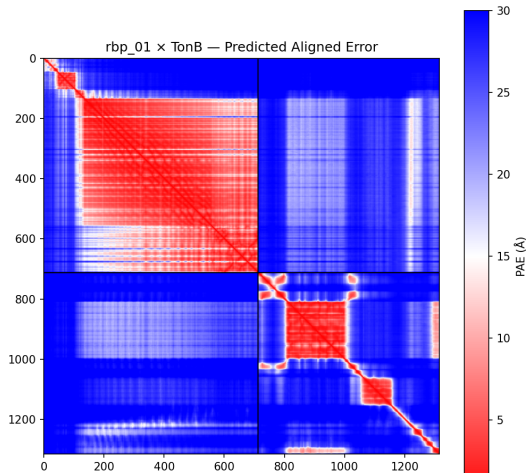
History note: an earlier run (job 59949) used the wrong protein —a 85 aa P25 ORF mis-labeled as rbp_01. **Job 59986 uses the correct 712 aa rbp_01** from 03_rbp_identification/outputs/EU717894.1_rbp_sequences.faa.

The three numbers that matter

Metric	Value	Interpretation
interface_ipTM	0.365	Low —model uncertain how they dock.
chain_A_ptm	0.808	High —rbp_01 monomer well-predicted.
confidence_score	0.683	Moderate overall complex quality.

The low ipTM **is not a failure** —it defines the experiment. That uncertainty IS what the ELISA + active learning loop is designed to answer. High chain A ptm means rbp_01 is structurally well-constrained → reliable for variant design.

PAE heatmap —interface block



Where the Boltz-2 outputs live

```
05_structure_prediction/outputs/boltz2/  
EU717894.1_rbp_01__GCF_000007145.1_tonB/
```

Inside that directory:

- `affinity.json` —interface_ipTM, chain_A_ptm, confidence_score.
- `boltz_results_.../predictions/.../*.pdb` —full atomic coordinates (open in PyMOL / ChimeraX).
- `pae_*.npz` —PAE matrix.
- `plddt_*.npz` —per-residue pLDDT (range 0.30-0.98, mean 0.76).

Full paths in `docs/planning/PI_briefing_2026-05-11.md` (no abbreviations).

Boltz-2 reminder: `predicted_dG = null` because the affinity head is small-molecule only.

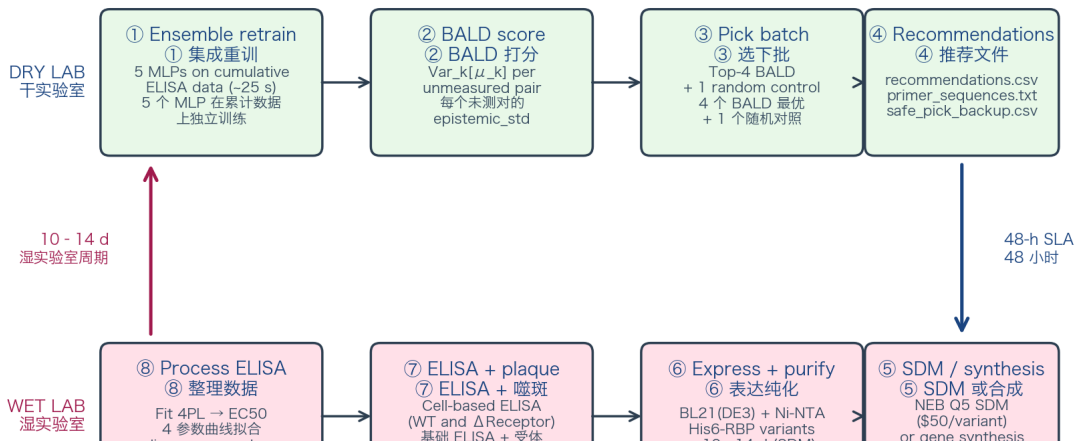
Part 5 —The 48-hour cycle · [WET LAB] [DRY LAB]

Key vocabulary —for this part

- **Gibson assembly** —one-pot reaction (5' exonuclease + polymerase + ligase) that joins overlapping DNA fragments into a circular plasmid. Cycle 0 cloning.
- **SDM (site-directed mutagenesis)** —a single targeted change on an existing plasmid (e.g., K450A). ~\$50 / 4 days; what we use Cycle 1+.
- **BL21(DE3) / pET-28a / IPTG** —E. coli expression strain + plasmid backbone + the inducer that flips on the gene.
- **His6 + Ni-NTA** —six histidines on the N-terminus stick to a nickel column → one-step purification (IMAC).
- **SDS-PAGE** —gel electrophoresis that separates proteins by size; verifies purity.
- **EC50 / 4PL fit** —half-max concentration from a 4-parameter logistic fit on the binding curve. Our quantitative “binding strength” —the model’s target variable.
- **Plaque / PFU** —a clear spot on a bacterial lawn = one infectious phage; PFU/mL = stock potency.
- **pK18mobsacB / sacB selection** —suicide plasmid for clean gene deletions in *Xanthomonas*; **sacB** makes cells die in sucrose, so we can select cells that lost the plasmid via a second crossover.
- **Electroporation** —brief electric pulse to permeabilise bacterial walls so plasmid DNA can enter.

Cycle structure

The 48-hour Active-Learning Cycle / 48 小时主动学习循环



Wet lab → dry lab handoff

End of each cycle, wet lab delivers to `08_cycle_data/outputs/cycle_<N>/`:

File	Required columns
<code>elisa_processed.csv</code>	<code>variant_id</code> , <code>receptor_id</code> , <code>ec50_nM</code> , <code>hill_slope</code> , <code>r2</code> , <code>plate_id</code> , <code>date</code>
<code>plaque_results.csv</code>	<code>variant_id</code> , <code>strain_id</code> , <code>pfu_per_ml</code> , <code>date</code>
<code>qc_report.md</code>	SDS-PAGE image path, concentration, expression notes

Minimum to retrain: ≥ 3 valid EC50 per variant with $R^2 > 0.9$. Failed variants marked `ec50_nM = NaN + failed_reason` —the model handles missing data.

Dry lab → wet lab handoff (48-hour SLA)

After receiving ELISA data:

File	Purpose
<code>recommendations.csv</code>	4 BALD + 1 random; primary task list
<code>primer_sequences.txt</code>	NEB Q5-compatible SDM primers
<code>uncertainty_bands.png</code>	Calibration: predicted vs measured (previous cycle)
<code>safe_pick_backup.csv</code>	Top-10 BALD, used only if 48-h SLA misses
<code>run_meta.json</code>	Git SHA + timestamp + pool stats (provenance)

Wet lab **does not know** which row is the random control —preserves blinded baseline for retrospective AL vs random comparison at project end.

Cloning execution

- **Cycle 0** —gene synthesis (IDT/Twist), 4-6 variants from structure-based design. ~2 weeks lead time, ~\$150/variant.
- **Cycle 1+** —site-directed mutagenesis (NEB Q5), point mutations on existing constructs. ~4 days, ~\$50/variant ($3\times$ cheaper, $3.5\times$ faster).
- BALD on a small data regime selects point mutations → SDM is the natural execution method.

Vector: pET-28a (Addgene 69864-3). Host: BL21 (DE3). Induction: 0.5 mM IPTG, 18 °C overnight (favors soluble trimer assembly).

ELISA protocol —cell-based binding

Adapted from Boeckaerts 2024 + Latka 2021:

- 1 Coat 96-well plate with heat-inactivated *Xanthomonas* (10^8 CFU/well).
- 2 Block 3 % BSA, 1 h.
- 3 Serial dilution His6-RBP (1 nM - 10^{-8} M).
- 4 HRP-anti-His6 detection, TMB substrate, read OD₄₅₀.
- 5 **4PL fit** → **EC50** is the active-learning target variable.

Controls per plate: BSA-only (background); WT-RBP (inter-plate normalizer); heat-denatured RBP (folding-specific binding); 3 technical replicates.

Receptor knockouts —pK18mobsacB (Schäfer 1994)

Markerless deletion via suicide vector + sucrose counter-selection:

- 1 Build deletion plasmid: ~500 bp upstream + ~500 bp downstream + pK18mobsacB.
- 2 Electroporate Xcc isolate; select on kanamycin (single crossover).
- 3 Counter-select on sucrose (sacB kills non-resolvers); PCR-confirm.

Targets, from Hung 2003:

- Δ tonB, Δ exbB, Δ exbD1 → expected to **block** phiL7 infection.
- Δ exbD2 → expected to **retain** infection (built-in negative control).

Timeline: 4-6 weeks per gene (parallelizable).

Validation tiers

Tier	What you measure	Story
1	ELISA only (WT host)	" We found variants that bind better."
2	+ Plaque assay (WT)	" Binding → infection confirmed."
3	+ Δ tonB / Δ exbB / Δ exbD1	" Receptor-specific causality." (paper-grade)

Recommendation (PI briefing): commit to Tier 3 if knockouts start May 17. Δ exbD2 negative control validates the entire knockout system for free.

Quality gates and failure modes

Failure	Action
Dry lab misses 48-h SLA	Wet lab uses <code>safe_pick_backup.csv</code>
Variant insoluble	Mark NaN; attempt backup truncation
ELISA $R^2 < 0.9$	Down-weight in retraining
All 5 picks fail QC	1 expert pick + 2 random; retrain on partial data
Calibration ECE > 0.1	Apply temperature scaling before next BALD

Part 6 —Reproducing & demoing · [DRY LAB] [WET LAB]

Key vocabulary —for this part

- **Branch** —parallel line of git history. All code on `active-learning-pipeline`; `main` is intentionally empty.
- **Conda env** —pinned-version Python install. `conda activate igem2026` before anything else, or imports will fail.
- **pytest** —Python' s standard test runner. Each module ships a small suite; passing = contract intact.
- **JupyterLab** —browser-based interface for `.ipynb` notebooks.
- **Laguna (CARC)** —USC' s GPU cluster. For jobs too big for a laptop (Boltz-2, ESM-2 650M, AF3).
- **SLURM** —Laguna' s job scheduler. `sbatch script.slurm` submits a job to a free GPU node.
- **CUDA** —Nvidia' s GPU compute platform. We pin `CUDA 12.1 / torch==2.5.1+cu121` to match the cluster driver.

Quick-start

```
# Clone + branch / 克隆并切换分支
git clone https://github.com/Alex-Ch3n/iGEM_Claremont_2026.git
cd iGEM_Claremont_2026
git checkout active-learning-pipeline

# Environment (one-time)
conda env create -f shared/env/environment.yml
conda activate igem2026
pip install nbstripout && nbstripout --install

# Minimum genomes for local dev (~5 MB total)
python 00_raw_data/processes/fetch_phages.py --accession EU717894.1
python 00_raw_data/processes/fetch_phages.py --accession NC_001604.1
python 00_raw_data/processes/fetch_bacteria.py --accession GCF_000007145.1

# Launch JupyterLab
jupyter lab
```

Full per-module checklists: GETTING_STARTED.md.

Per-module entry points

Module	Open this first
00	00_raw_data/processes/01_verify_dataset.ipynb
01	01_data_ground_truth/processes/01_fetch_reference_genomes.ipynb
02	02_annotation/processes/01_run_phanotate.ipynb
03	03_rbp_identification/processes/01_run_phagerbpdetect.ipynb
04	04_protein_embedding/processes/01_embed_esm2.ipynb
05	05_structure_prediction/processes/01_run_boltz2.ipynb (Laguna for production)
06	06_uncertainty_model/processes/run_cycle0.py
07	07_acquisition_function/processes/run_bald.py

Where outputs live

File	Path (from repo root)
Interaction matrix	01_data_ground_truth/outputs/interaction_matrix.csv
RBP candidates	03_rbp_identification/outputs/EU717894.1_rbp_candidates.csv
ESM-2 embeddings	04_protein_embedding/outputs/embeddings_esm2_t6_8M_phiL7_rbps.npz
Boltz-2 PDB / ipTM	05_structure_prediction/outputs/boltz2/EU717894.1_rbp_01__GCF_000007145.1_tonB/
Ensemble predictions	06_uncertainty_model/outputs/cycle_0/predictions.csv
Calibration plot	06_uncertainty_model/outputs/cycle_0/calibration.png
BALD recommendations	07_acquisition_function/outputs/cycle_1/recommendations.csv
PAE heatmap	pae_heatmap.png (repo root)

Live demo plan (full runbook → DEMO.md)

- 1 **Module 03** —`pytest 03_rbp_identification/processes/tests/ -v` (~15 s).
- 2 **Module 04** —inspect cached `.npz` (shape (3, 320); seq_ids match rbp_01/02/03) (<1 s).
- 3 **Module 06** —`python 06_uncertainty_model/processes/run_cycle0.py` (~3 s, verified).
- 4 **Module 07** —`python 07_acquisition_function/processes/run_bald.py --cycle 1 --n_bald 4 --n_random 1` (<1 s).
- 5 **Module 05 (read-only)** —open `pae_heatmap.png` and the PDB in PyMOL.

Each step has “what to say” + “if it fails” in DEMO.md.

Laguna HPC —when to push to GPU

Task	Local?	Why Laguna?
ESM-2 8M embedding (3 RBPs)	Yes	seconds on CPU
ESM-2 650M / 3B (777 phages)	No	GPU memory + time
Boltz-2 protein-protein	No	15 min/pair, A100/L40S only
AF3 batch	No	GPU + weights approval
Ensemble train + BALD	Yes	seconds on CPU

Setup recipe + SLURM templates: `LAGUNA.md` (CUDA 12.4, torch 2.5.1+cu121).

Tests —current pass rates

Module	Tests
00 Raw Data	15+ pass (3 expected fails —GCF/T7 not in seed list)
01 Ground Truth	22/22
02 Annotation	26/26
03 RBP ID	25+ pass (2 expected —HMM needs local hmmpress)
04 Embedding	17/17
05 Structure	28/28 (1 expected skip —needs completed GPU run)
06 Uncertainty Model	9/9
07 BALD	18/18

Part 7 —Risks, decisions, asks · [PI]

Pending PI decisions

Decision	Deadline	Default
pK18mobsacB vs CRISPRi for knockouts	May 17	pK18mobsacB (Hung 2003 used it in Xcc)
AlphaFold 3 weights —who applies?	This week	Either Alex or PI; institutional email preferred
Validation tier commitment	Before June 1	Tier 3 if knockouts start May 17
Phage enrichment source (LA County)	Before June 1	TBD —needs PI input on lab access
Manuscript ambition	Before Cycle 0	Concurrent submission Bioinformatics / NAR Genomics

Critical risks + mitigations

Risk	Likelihood / Impact	Mitigation
Strain isolation fails	L / H	Dual source (brassica + citrus); 2 sampling rounds
Phage isolation yields nothing lytic	M / H	Sewage / runoff enrichment; Phage Directory backup
RBP expression insoluble	M / M	T7 gp17 positive control first; GCN4 trimer tag backup
ELISA dynamic range insufficient	M / H	2 weeks of Cycle 0 reserved for ELISA optimization
AL underperforms random	L / H (reportable)	Standard control arm; honest reporting
Dry lab misses 48-h SLA	M / M	<code>safe_pick_backup.csv</code> pre-approved by PI
Receptor knockout fails	M / M	Multiple targets in parallel; CRISPRi fallback

What “AL underperforms random” looks like

- After Cycle 2: BALD trajectory's test R^2 **statistically equivalent** to random.
- We commit to honest reporting:
 - Document the negative result with the same rigor as a positive.
 - Useful contribution to the field —Hie 2024 worked; ALDE worked; phage-RBP didn't (yet).
- Concrete metrics archived per cycle:
 - Test-set R^2 on held-out variants.
 - Calibration ECE.
 - Information gain per experiment (KL divergence posterior vs prior).

Timeline

Date	Milestone
2026-05-17	Wet lab launches: brassica sampling, pK18mobsacB construction, Cycle 0 synthesis order
2026-06-01	Cycle 0 starts: ELISA optimization + first binding measurements
2026-06-14	Cycle 1: ensemble retrained, first BALD picks delivered, SDM
2026-06-28	Cycle 2: round-2 recommendations, receptor knockouts complete
2026-07-12	Analysis + extension: trained model on soil-derived phage genomes
2026-09-12	Wiki freeze prep
2026-10-21	Wiki freeze
2026-11-13	Grand Jamboree

iGEM deliverables checklist

- **Wiki:** Engineering DBTL · Modeling · Hardware/Software · Human Practices.
- **Composite Part:** ≥ 4 -6 RBP-His6 constructs from in-house phage isolates.
- **Software repo:** open-source closed-loop AL pipeline (this repo).
- **Promotion video:** demo of one closed-loop cycle.
- **Community contribution:** in-house *Xanthomonas* + phage isolates → iGEM Registry; sequenced genomes → NCBI.

Open questions for the team

- 1 Wet lab —what' s the LA County sewage / agricultural runoff access?
- 2 Wet lab —do we want a positive expression control (T7 gp17) before committing the variant library?
- 3 Dry lab —who owns the ESM-2 3B Laguna job (1280→2560-D, final benchmark)?
- 4 Everyone —manuscript submission timing relative to iGEM wiki freeze?
- 5 PI —APHIS permit posture if our isolate ID reveals a quarantine-status pathovar.

Part 8 —References & appendix

Key papers per module —compact

Module	Paper	Role
00 / 01	da Silva 2002 Nature; Lee 2009 AEM; Hung 2003 BBRC	reference genomes
02	McNair 2019 Bioinformatics (PHANOTATE); Hyatt 2010 (Prodigal); Bouras 2023 (pharokka)	gene calling
03	Boeckaerts 2022 Viruses (PhageRBPdetect)	RBP HMM + XGBoost
04	Lin 2023 Science (ESM-2); Liu 2025 Nat Commun (PLM-interact)	embeddings + PPI prediction
05	Abramson 2024 Nature (AF3); Passaro 2025 (Boltz-2)	structure + ipTM
06	Lakshminarayanan 2017 NeurIPS; Greenman 2025 PLoS Comput Biol	deep ensembles + UMAP
07	Houlsby 2011 (BALD); Yang 2025 Nat Commun (ALDE); Hie 2024 Nat Biotechnol	acquisition
08	Schäfer 1994 (pK18mobsacB); Gibson 2009; Latka 2021 mBio	wet lab methods

Full annotated reading guide: [docs/reference/papers.md](#) (19 papers).

Five literature corrections (May 2026 audit)

Was	Corrected to
" TonB-ExbB-ExbD1-ExbD2 all essential" (Wang 2003)	TonB-ExbB-ExbD1 essential; ExbD2 NOT (Hung 2003 BBRC)
" Boltz-2 affinity head for RBP × receptor"	affinity head is small-molecule only → NaN ; use ipTM
" Greenman 2025 NAR Genomics; deep ensemble = best UQ"	Greenman 2025 PLoS Comput Biol; " no single best UQ"
" Hie 2024 used ESM-2, ~50 antibody variants"	Hie 2024 used ESM-1b/1v, ~20 variants per antibody
" Lee 2009 named the tail spike"	Lee 2009 searched and found no OP1 ORF25 homolog

Full audit: `docs/reference/paper_reading_notes.md`.

Navigation reference

Want to know...	Read this
What each module produces	<module>/README.md
End-to-end status snapshot	docs/planning/PI_briefing_2026-05-11.md
Project plan (EN, full)	docs/planning/iGEM_2026_Project_Plan.md
Project plan (ZH, full)	docs/planning/iGEM_2026_ 项目大纲 _ 中文版.md
Paper list (annotated)	docs/reference/papers.md
Literature corrections	docs/reference/paper_reading_notes.md
Data contracts	INTERFACE.md
HPC setup	LAGUNA.md
Wet-lab SOPs	docs/protocols/ (5 Benchling PDFs)

Glossary —fast lookup

Term	Meaning
RBP	Receptor-binding protein —the phage’ s ” key” to the host cell
HMM	Hidden Markov Model —sequence-profile method (Boeckaerts 2022)
ESM-2	Evolutionary-scale protein language model (Lin 2023)
ipTM	Interface predicted TM-score (0-1, structural confidence)
ELISA	Enzyme-linked immunosorbent assay (binding readout)
EC50	Half-maximal effective concentration (from 4PL fit)
SDM	Site-directed mutagenesis (NEB Q5 kit)
BALD	Bayesian Active Learning by Disagreement (Houlsby 2011)
Epistemic	Reducible model uncertainty (= BALD target)
Aleatoric	Irreducible measurement noise

Full glossary: <docs/reference/glossary.md>.

Citation list —receptor biology + Xanthomonas pathology

- 1 **da Silva 2002** —Nature 417:459-463. Xcc ATCC 33913 reference genome (GCF_000007145.1 / GenBank AE008922).
- 2 **Lee 2009** —Appl Environ Microbiol 75(24):7828-7838. phiL7 genome (NCBI EU717894). Audit: actively searched for and could not find an OP1 ORF25 tail-fiber homolog.
- 3 **Hung 2003** —BBRC 302(4):878-884, PMID 12646254. TonB / ExbB / ExbD1 essential for phiL7; ExbD2 NOT required. Audit: the previously-cited “Wang 2003” was a hallucinated reference.
- 4 **Ryan 2011** —Nat Rev Microbiol 9:344-355. >400-species host range; economic context.
- 5 **Aiello 2019** —Plant Dis. Copper resistance in Xanthomonas spp.
- 6 **Iriarte 2018** —Front Plant Sci 9. Phage biocontrol field trial on tomato.
- 7 **Holtappels 2022** —Microb Biotechnol 15(3):597-610. Phage biocontrol in integrated plant protection.
- 8 **Farquharson 2021** —T4 × E. coli: binding ≠ infection (RBP binds 85 % of strains, plaques on 11 %).

Citation list —host-range prediction + annotation

- 9 Boeckaerts 2022** —Viruses 14(6):1329. PhageRBPdetect HMM + XGBoost (precision-recall AUC 93.8 %, F1 84.0 %).
- 10 Boeckaerts 2024** —Nat Commun 15:4768/48675. PhageHostLearn (strain-level). Audit: AUC “0.82” = up to 81.8 % at 100 % identity threshold; cross-strain ~0.70.
- 11 Mutalik 2025** —bioRxiv (preprint). PAML benchmark: cross-genus AUC 0.67-0.70.
- 12 McNair 2019** —Bioinformatics 35(22):4537-4542. PHANOTATE (dynamic-programming phage ORF caller).
- 13 Hyatt 2010** —BMC Bioinformatics 11:119. Prodigal (used via pyrodigal).
- 14 Bouras 2023** —Bioinformatics 39(1):btac776. Pharokka phage annotation pipeline.

Citation list —protein language models, structure, UQ

- 15 Lin 2023** —Science 379(6637):1123-1130. ESM-2 (~65 M sequences; 650M → 1280-D, 3B → 2560-D).
- 16 Liu 2025** —Nat Commun 16:64512, DOI 10.1038/s41467-025-64512-w. PLM-interact PPI transfer (+16-28 % AUPR). Audit: untested on phage-bacteria.
- 17 Abramson 2024** —Nature 630:493-500, DOI 10.1038/s41586-024-07487-w. AlphaFold 3.
- 18 Passaro 2025** —bioRxiv (preprint). Boltz-2. Audit: affinity head is small-molecule only (PubChem/ChEMBL/BindingDB); outputs NaN for protein-protein pairs. Use ipTM as structural confidence proxy.
- 19 Lakshminarayanan 2017** —NeurIPS 30, arXiv:1612.01474. Deep ensembles + Gaussian NLL.
- 20 Ovadia 2019** —NeurIPS 32. UQ under dataset shift.
- 21 Greenman 2025** —PLoS Comput Biol 21(1):e1012639, DOI 10.1371/journal.pcbi.1012639. Audit: journal is PLoS Comput Biol (NOT NAR Genomics); conclusion: “no single best UQ method.”

Citation list —active learning + wet lab methods

- 22 Lindley 1956** —Ann Math Stat 27:986-1005. Bayesian optimal experimental design.
- 23 Settles 2009** —CS Tech Rep 1648, U. Wisconsin-Madison. Active Learning Literature Survey.
- 24 Houlisby 2011** —arXiv:1112.5745. BALD. Audit: original GPC classification; deep-ensemble regression is our extension.
- 25 Hie 2024** —Nat Biotechnol 42:275-283 (preprinted 2022). Audit: ESM-1b/1v (NOT ESM-2); ~20 variants per antibody; language-model-likelihood filtering, not BALD closed loop. Earlier docs that cite “Hie 2022 Cell” refer to this same paper.
- 26 Yang 2025** —Nat Commun 16:55987, DOI 10.1038/s41467-025-55987-8. ALDE. Audit: uses Thompson sampling + one-hot (NOT BALD + ESM-2).
- 27 Wittmann 2021** —Cell Syst 12(11):1026-1045. ML-assisted directed evolution; exploration vs exploitation.
- 28 Scherfer 1994** —Gene 145(1):69-73. pK18mobsacB (Addgene #87097).
- 29 Gibson 2009** —Nat Methods 6:343-345. Gibson assembly.
- 30 Studier & Moffatt 1986** —J Mol Biol 189(1):113-130. pET / BL21(DE3) expression system.
- 31 Latka 2021** —mBio 12(6):e02329-21. RBP modular truncation. Audit: depolymerase mechanism; structural principle transfers, biochemistry doesn't.

Document log

Date	Change
2026-05-17	Initial draft: ~71 slides + guide + DEMO runbook + 4 figures. Pandoc + Beamer + xelatex + xeCJK / Hiragino Sans GB.
2026-05-17	Path-verification pass; added Module 04 to demo plan (slide 48 + DEMO.md); corrected phage_genomes/ → phage/<acc>/; HMM setup via setup_inputs.sh. Committed as 5304f69.
2026-05-18	Added “Key vocabulary” slide at the start of Parts 1-6 (Part 3 gets two —neural-net basics + UQ/AL). Aligned with the equivalent guide_en.md callouts.
2026-05-18	Drive-by fix: “Hie 2022” → “Hie 2024” (same paper). Added the 4-slide citation appendix (31 entries with audit notes) and this log.

Author for all entries: Claude (onboarding agent), reviewed by Alex Chen. Fuller version of this log lives in docs/onboarding/guide_en.md.

Thank you / 谢谢

Questions, comments, push-back welcome.

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- PI: Prof. J. Cesar Ignacio-Espinoza
- Faculty advisor: Prof. Ran Libeskind-Hadas
- Wet lab leads: Sarah, Olivia, Weitao, Carol
- Contributors: Ryan, Leah

Live demo (15 min) follows. See `docs/onboarding/DEMO.md`.